Alexander Martin

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EDUCATION

Johns Hopkins University — **Baltimore, Maryland**
Ph.D. in Computer Science — Expected 2028
Advisor: Dr. Benjamin Van Durme

University of Rochester — **Rochester, New York**
B.S. in Computer Science; Highest Honors in Research — May 2024
Advisor: Dr. Aaron Steven White, Dr. Jiebo Luo
Thesis: Human-Centric Event Representations in Documents and Videos

SELECT HONORS & AWARDS

National Science Foundation, **Graduate Research Fellowship** — 2024 – 2029
Computing Research Association, **Outstanding Undergraduate Research Award (HM)** — 2024
University of Rochester, **Charles L. Newton Prize** — 2024

INDUSTRY RESEARCH

Amazon, Fire TV — 2025 – Present
Researcher; AI2AI Award
Researched topics in representations for efficient multimodal retrieval and generation.

RESEARCH EXPERIENCE

Human Language Technology Center of Excellence — 2024 – Present
Ph.D. Researcher; Advised by Dr. Benjamin Van Durme
Researched topics in multimodal content generation and reasoning using documents and videos.

Human Language Technology Center of Excellence — Summer 2024
Researcher Intern; Advised by Dr. Benjamin Van Durme
Researched extracting information about events from videos and aligned text and video retrieval.

Formal And Computational Semantics Lab — 2022 – 2024
Undergraduate Researcher; Advised by Dr. Aaron Steven White
Researched extracting and summarizing information about events from large unstructured text.

Visual Intelligence & Social Multimedia Analytics Lab — 2022 – 2024
Undergraduate Researcher; Advised by Dr. Jiebo Luo
Researched methods for image generation, image-to-image translation, and video understanding.

Environmental Protection Agency — Summer 2022 – Fall 2022
Research Intern; Advised by Dr. Andrea Kirk
Developed methods for relative importance analysis to measure the effects of PFAS exposure on humans and their health, including cancer risk and bone mineral density.

Rochester Human Computer Interaction Lab — 2022 – 2024
Research Assistant; Advised by Dr. Ehsan Hoque
Created synthetic datasets to improve performance of hand pose estimation models for diagnosing Parkinson's Disease in virtual health appointments.

PUBLICATIONS

Selected Publications: Multimodal RAG, Reasoning, Efficiency

[1] H. Qin*, **A. Martin***, et al. "Multi-Vector Index Compression in Any Modality" (ArXiv 2026)
[2] T. Skow*, **A. Martin***, et al. "RANKVIDEO: Reasoning Reranking for Text-to-Video Retrieval" (ArXiv 2026)
[3] **A. Martin**, et al. "*WikiVideo: Article Generation from Multiple Videos*" (ArXiv 2025)
[4] **A. Martin**, et al. "*Seeing Through the MiRAGE: Evaluating Multimodal Retrieval Augmented Generation*" (ArXiv 2025)

[5] A. Reddy*, **A. Martin***, et al. "*Video-ColBERT: Contextualized Late Interaction for Text-to-Video Retrieval*" (CVPR 2025)

All others: Google Scholar

TALKS

Knowledge Intensive Multimodal Retrieval Augmented Generation [talk] **November 2025**
University of Maryland, Baltimore County; PPR Seminar Seminar

Discovering and Understanding Events in Videos in Real Time [slides] **October 2025**
MacroXStudio (macrox.ai) Talk

On Demand Article Generation of Events in Real Time [slides] **April 2025**
University of Rochester, Department of Computer Science Seminar

MARE: Automatic Modality-Agnostic Report Evaluation [slides] **April 2025**
Eval4Rag Workshop; European Conference on Information Retrieval Oral

Understanding Events in Multimodal Data Through Question Answering [slides] **April 2025**
4th Workshop on Processing and Evaluating Event Representations Talk

FAMuS: Frames Across Multiple Sources [slides] **April 2024**
3rd Workshop on Processing and Evaluating Event Representations Talk

Jurassic World Remake [slides] **October 2023**
Main Conference; ACM Multimedia Ora

SERVICE

MAGMaR: Workshop on Multimodal Augmented Generation via Multimodal Retrieval
Organizing Committee; *ACL 2026*
Program Committee; *ACL 2025*

Mentorship
Charles Weng (**Initial Placement:** TikTok Research Scientist)
Dengjia Zhang (**current:** Johns Hopkins University, M.S. 2026)
Hanxiang Qin (**current:** Johns Hopkins University, M.S. 2026)
Katherine Guerrerio (**current:** Johns Hopkins University, B.S. 2027)

Community Service
North American Computational Lingusitics Open Competition (NACLO) 2024 – Present
Competition Organizer
STEM Initiative 2021 – 2024
Education Mentor
University of Rochester: Residential Life 2021 – 2024
Resident Advisor

Teaching
Introduction to Artificial Intelligence (University of Rochester) Spring 2023
Data Structures and Algorithms (University of Rochester) Spring 2022, Fall 2022
Introduction to Computer Science (University of Rochester) Fall 2021

OTHER HONORS & AWARDS

University of Rochester, **Senior Research Award** 2024
University of Rochester, **Research Presentation Grant (x2)** 2023, 2024
University of Rochester, **Dean's Award in Engineering and Mathematics (x2)** 2023, 2024
University of Rochester, **River Campus Libraries Dataset Grant** 2023
University of Rochester, **Residential Life Best Program of the Year** 2023
University of Rochester, **Make It Happen Grant** 2022